

QIANQIAN WAN

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EDUCATION

The Ohio State University

Columbus, OH, USA

Ph.D. in Psychology

08/2020–12/2024

- Dissertation: *Unraveling Cognitive Processes Driving the Development of Optimized Category Learning and Decision-Making*
- Advisor: Vladimir Sloutsky, Ph.D.
- Committee: Brandon Turner, John Opfer, Hsin-Hung Li

University of Hong Kong

Hong Kong

B.Soc.Sc. in Psychology

09/2016–05/2020

- Honors Thesis: *Frames of Reference for Audiovisual Bounce-Inducing Effect*
- Advisor: Dorita H. F. Chang, Ph.D.

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

- Wan, Q., & Sloutsky, V. M. (2026). Working memory shapes information sampling and attention allocation across development. *Journal of Experimental Psychology: General*, 155(2), 479–498. <https://doi.org/10.1037/xge0000000>
- Wan, Q., & Sloutsky, V. M. (2026). Development of uncertainty sensitivity: Early competence or protracted development? *Developmental Science*, 29(2), e70137. <https://doi.org/10.1111/desc.70137>
- Wan, Q., Savic, O., Gao, M., Ralston, R., O’Leary, A., & Sloutsky, V. (2025). From uncertainty to performance optimization: Longitudinal insights into metacognitive development. *Child Development*. Advance online publication. <https://doi.org/10.1111/cdev.70029>
- Semizer, Y., Yu, D., Wan, Q., Balas, B., & Rosenholtz, R. (2025). Effects of maze appearance on maze solving. *Attention, Perception, and Psychophysics*, 87(2), 637–649. <https://doi.org/10.3758/s13414-024-03000-7>
- Wan, Q., & Sloutsky, V. (2024). Exploration, distributed attention, and development of category learning. *Psychological Science*, 35(10), 1164–1177. <https://doi.org/10.1177/09567976241258146>

UNDER REVIEW

- Wan, Q., Moriarty, N., Duarte, S. E., Ghetti, S., & Geng, J. J. (under review). The development of navigation and attentional abilities shapes how children explore naturalistic environments.

RESEARCH EXPERIENCE

Center for Mind and Brain & Center for Neuroscience, UC Davis

Davis, CA, USA

Postdoctoral Researcher

01/2025–present

- Leading a multi-session fMRI study examining how goal-directed attention during navigation shapes spatial learning

- Developed immersive VR paradigm tracking body, head, and eye movements during naturalistic exploration in children and adults
- Mentors: Charan Ranganath, Ph.D. & Joy Geng, Ph.D.

The Ohio State University
Graduate Researcher

Columbus, OH, USA
08/2020–12/2024

- Designed novel behavioral paradigms investigating attention, working memory, and category learning
- Published findings in *Psychological Science* and *Journal of Experimental Psychology: General*
- Collected behavioral data from 300+ child participants across developmental studies
- Developed computational models of attention allocation during learning
- Mentor: Vladimir Sloutsky, Ph.D.

Johns Hopkins University
Summer Research Assistant

Baltimore, MD, USA
06/2019–09/2019

- Investigated perceptual mechanisms underlying fire perception
- Mentors: Chaz Firestone, Ph.D. & Justin Halberda, Ph.D.
- Funded by Summer Research Fellowship

Massachusetts Institute of Technology
Summer Research Assistant

Cambridge, MA, USA
07/2018–09/2018

- Examined perceptual factors in maze solving using computational approaches
- Mentor: Ruth Rosenholtz, Ph.D.

University of Hong Kong
Undergraduate Researcher

Hong Kong
11/2018–05/2020

- Investigated audiovisual bounce-inducing effect using VR to manipulate gravity perception
- Mentor: Dorita H. F. Chang, Ph.D.

TECHNICAL SKILLS & METHODS

Neuroimaging: fMRI (design, acquisition, representational similarity analysis), EEG

Behavioral Methods: Immersive VR paradigm development (Unity), eye-tracking, mobile behavioral assessment, body and head motion tracking

Programming: Python, R

Statistical Methods: Mixed-effects modeling, Bayesian analysis, computational modeling of attention and learning

TEACHING EXPERIENCE

The Ohio State University
Instructor

Columbus, OH, USA
08/2023–12/2024

- Introduction to Psychology: Designed weekly in-class experiments; students participated in demonstrations before learning theoretical frameworks

The Ohio State University
Teaching Assistant

Columbus, OH, USA
08/2021–05/2023

- Courses: Lifespan Development, Adolescence, Psychology of Adjustment, Psychology of Motivation, Psychology of Developmental Disabilities, Psychology of Childhood, Cognitive Aging Neurodegeneration and Neuroplasticity

MENTORING

HONORS THESIS SUPERVISION

- Sneha Prabu, *The Ohio State University* 05/2023–03/2024
- Karys DeWit, *The Ohio State University* 05/2024–present
- Krisha Patel, *The Ohio State University* 05/2024–01/2025
- Cameron Wallace, *The Ohio State University* 01/2025–present

RESEARCH ASSISTANTS

- Carley Smith, *UC Davis* 01/2025–08/2025
- Alexdeep Grewal, *UC Davis* 01/2025–11/2025
- Prati Mehta, *UC Davis* 08/2025–present
- Sophie Hu, *UC Davis* 08/2025–present
- Aneesh Bhardwaj, *UC Davis* 01/2026–present

PRESENTATIONS

CONFERENCE TALKS

- Multimodal Attention Capture in Real-World Exploration: A Developmental VR Study of Body-Head-Eye Coordination, *Psychonomic Society* 11/2025
- How Working Memory Shapes Early Attention and Learning, *Society for Research in Child Development* 05/2025
- Driven by Information: Children’s Exploration Shapes Their Distributed Attention in Category Learning, *Cognitive Science Society* 07/2023
- A Longitudinal Investigation on Explore-Exploit Dilemma, *European Society for Cognitive Psychology* 09/2023

CONFERENCE POSTERS

- Developmental Differences in a Naturalistic Exploration Task, *Vision Science Society; Bay Area Developmental Symposium* 2026
- A Causal Link between Working Memory Capacity and Attention Distribution in Category Learning, *Cognitive Science Society* 07/2024
- A longitudinal investigation of the early development of metacognitive monitoring and control, *Cognitive Development Society* 03/2022
- Verbal working memory capacity modulates category representation, *Cognitive Science Society* 07/2021
- Perceptual factors in mental maze solving (with Yu, D.), *Vision Science Society* 05/2019

HONORS & AWARDS

- University Fellowship, *The Ohio State University* 2020
- SBS Graduate Scholarship Award, *The Ohio State University* 2021
- Graduate Travel Grants, *The Ohio State University* 2021–2024

SERVICE

PROFESSIONAL SERVICE

- Conference Reviewer, *Cognitive Science Society* 2022–2025

DEPARTMENTAL & UNIVERSITY SERVICE

- Graduate Student DEI Committee, *Ohio State University* 2023–2024
- Organizer of Undergraduate Research Workshop, *Ohio State University* 2022–2024

COMMUNITY OUTREACH

- Scientific Ambassador, *Center of Science and Industry (COSI), Columbus, OH* 2022–2024

REFERENCES

Vladimir Sloutsky, Ph.D.

Professor of Psychology
The Ohio State University
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(614) 247-4449

Joy Geng, Ph.D.

Professor of Psychology
University of California, Davis
jgeng@ucdavis.edu
(530) 297-4486

Charan Ranganath, Ph.D.

Professor of Psychology
University of California, Davis
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(530) 757-8750